Visual Reconstruction with EEG

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Introduction

Individuals with motor disabilities due to paralysis, stroke, or other neurological diseases often have what is called *locked-in syndrome*, where they cannot communicate their thoughts or mental imagery, limiting their expression and quality of life.

Research groups have recreated images of what a person sees using their brain data (such as from fMRI) and machine learning models, referred to as *visual reconstruction*.

Electroencephalogram (EEG) is a non-invasive recording of electrical activity from the brain and is appealing because of its portability, prevalence, modest price, and high temporal resolution. However, prior work in EEG visual reconstruction has relied heavily on text, prioritizing object category and semantic meaning over specific visual features such as shape, color, orientation, and size. This has prevented visual reconstruction with EEG from producing realistic images.

Fig. 1 Reconstructions are all chairs and therefore *categorically correct*, but have *incorrect details* [1]

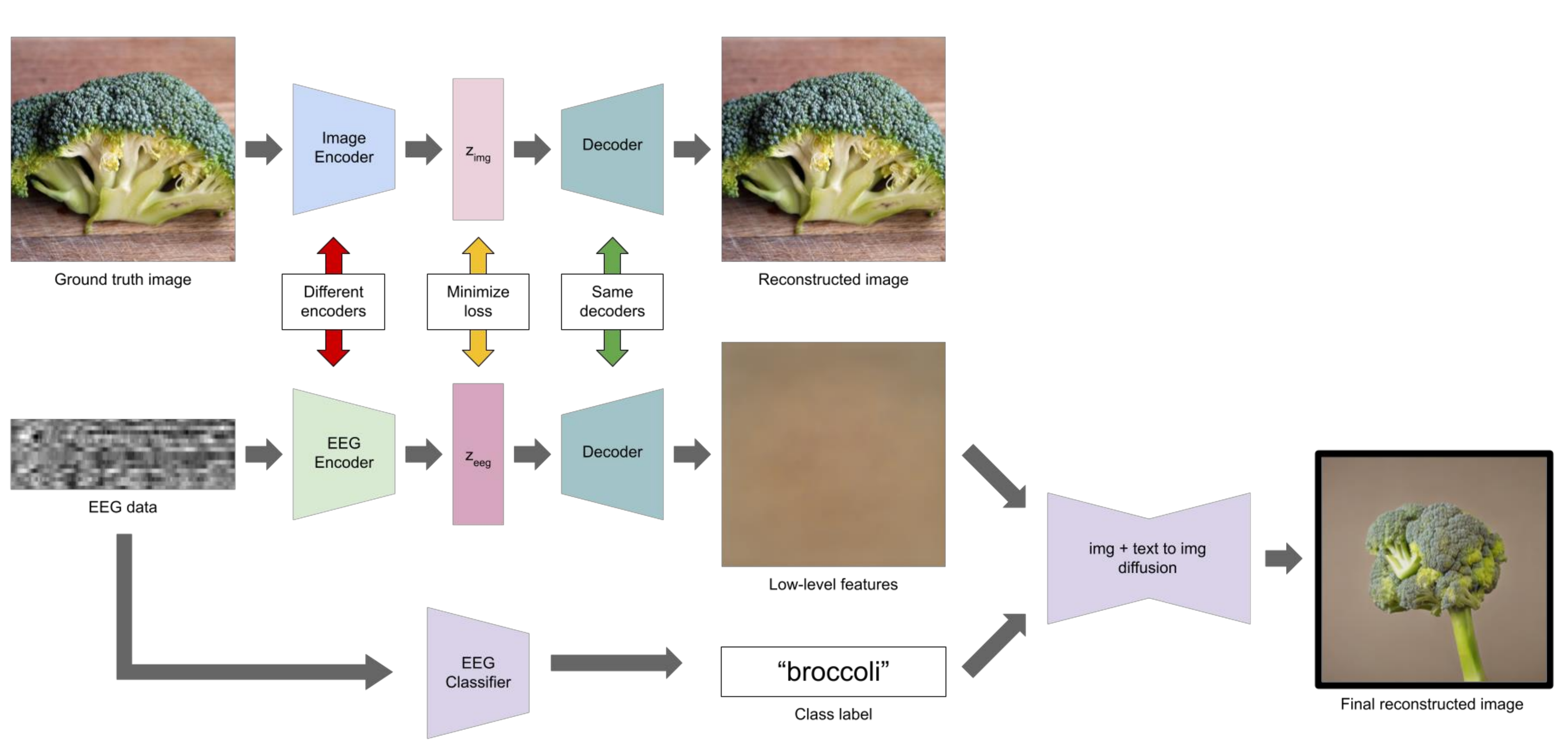

Fig. 2 Model architecture.

Goal

To create a machine learning model which prioritizes preserving specific visual features over object correctness, thereby producing true-to-experience visual constructions from EEG data.

Methods

Variational autoencoders (VAEs) are a type of machine learning model (artificial neural network) that takes in some input, encodes its important features in a latent space, and samples from this latent space to create an output as similar as possible to the input. We used an off-the-shelf, pre-trained VAE from StabilityAI (*stabilityai/sd-vae-ft-mse*) to first construct a latent representation of the images in our dataset.

We created an EEG encoder that generates low-level feature images when paired with Stable Diffusion's decoder. By minimizing the loss between the VAE and EEG latents, the model learns to represent EEG and image data in the same way, creating a model that takes in EEG data and outputs a reconstructed image of what the person saw. This reconstructed image only contains low-level features such as shape, color, and texture, due to EEG's low spatial resolution.

To improve the reconstruction, we also extracted a class label from the EEG data by creating a classifier. This label, along with the low-level features, was fed into CompVis's Stable Diffusion v1-4 to produce high-quality reconstructions.

The model was created using PyTorch. We used the publicly available EEG data from THINGS EEG-2, the largest, most robust and readily available EEG dataset.

Code

All code and data is publicly available in our Github repository, located at *github.com/IsaacVanB/visual-reconstruction-eeg.*

Results

For low VAE latent dimensions, the encoder was able to represent meaningful low-level information. Evaluated with the Structural Similarity Index Measure (SSIM), reconstructions from *low-level images + label* were more faithful to the ground truth image than reconstructions from *label only*. A one-sided paired permutation test showed that these improvements are statistically significant.

Despite the limitations of EEG data, we struck a balance between realistic and visually-accurate reconstructions.

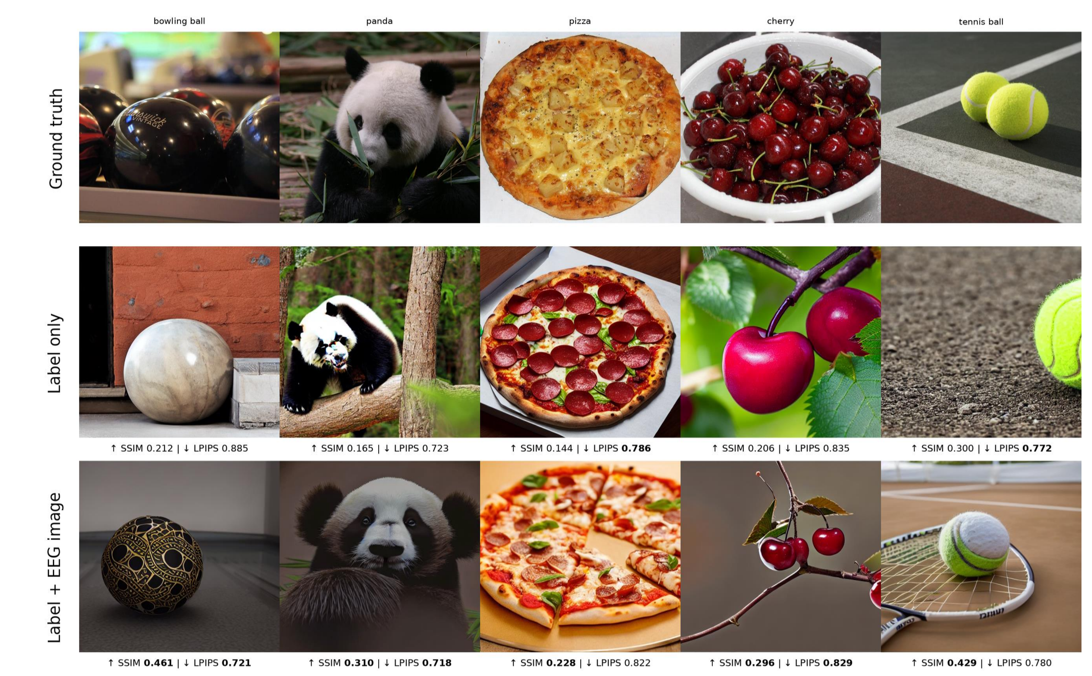

Fig. 3 Results. Reconstructed images from EEG dataset using our model. SSIM = 1 is perfect similarity.

Discussion

The project demonstrated that low-level visual features can be extracted from EEG data and used to improve reconstruction accuracy. However, EEG data lacks the spatial resolution to generate high-quality images without some form of semantic guidance. Combining EEG + fMRI is being explored by research groups, and could eliminate the need for diffusion. Our current model is also heavily reliant on the classifier, as Stable Diffusion will always generate an image, even if the class label is incorrect.

Aside from model-specific limitations, more general problems include:

- Small datasets
- Intersubject variability
- Real-time usage delay (for real-world implementation)

[1] *DreamDiffusion: Generating High-Quality Images from Brain EEG Signals*, 2023. https://arxiv.org/abs/2306.16934

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